

EduVision Programming

New Student Assessment Paper

Basic Information

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Current Year: 6

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Past experience learning C++ (terms): None

Past experience learning Python (terms): Basics and simple commands.

Part one --- Self-Assessment Trial Questions

These questions are optional. You are encouraged to attempt them. Please submit the source code together with the completed questionnaire.

Problem 1: Numbers Containing Digit 3 or 7

Given two integers m , n , find the number of integers from m to n (inclusive) which contain the digit 3 or 7.

Input Format

One line containing two integers m , n separated by spaces.

Output Format

Output one line containing one integer which is the number of integers from m to n that contain the digit 3 or 7.

Example

Input #1:

13 37

Output #1:

Explanation

Numbers satisfying the requirements are: 13, 17, 23, 27, 30, 31, 32, 33, 34, 35, 36, 37.

Constraints

$$1 \leq m \leq n \leq 10^6$$

$$0 \leq n - m \leq 10^7$$

Problem 2: Taxi Fare**Description**

A and B are two places d kilometers apart. Bob takes a taxi from A to B at h hours m minutes. Along the way, there are n checkpoints. Checkpoint i is s_i kms from A, the check takes t_i minutes. Taxi fare rules are:

Normal time (7:00–23:00): \$3 per km, \$2 per waiting minute.

Night time (23:00–7:00): \$5 per km, \$4 per waiting minute.

The taxi drives at a constant speed of 1 km per minute (ignoring acceleration and deceleration). Write a program to calculate how much Bob pays for the trip.

Input Format

1st line: four integers d, h, m, n , followed by n lines, each containing two integers s, t .

Output Format

One line, containing one integer which is the taxi fare.

Example

Input #1:

```
13 12 0 4
3 3
6 7
10 12
12 5
```

Output #1:

Constraints

$$0 \leq h < 24, 0 \leq m < 60$$

$$1 \leq n < d \leq 1000$$

$$1 \leq t_i \leq 100$$

$$\text{For any } 1 \leq i < j \leq n, 1 \leq s_i < s_j < d$$

Problem 3: Sum of All Expressions**Problem Description**

Given a string consisting of digits '1' to '9', you may insert the operator '+' between digits multiple times (or not at all) to form valid expressions. Calculate the sum of the values of all possible valid expressions.

Input Format

One line, containing a string s composed of digits '1' to '9'.

Output Format

One integer, which is the sum of the values of all valid expressions.

Example

Input #1:

125

Output:

176

Explanation #1:

All possible valid expressions are

125

$$1 + 25 = 26$$

$$12 + 5 = 17$$

$$1 + 2 + 5 = 8$$

Input #2:

9999999999

Output #2:

12656242944

Constraints

$1 \leq |s| \leq 10$

Part two ----- Circle your answer next to each question

Syntax and Standard Library

Input, Output, and Variables:

Not learned Learned **Proficient**

Conditional Branching:

Not learned **Learned** **Proficient**

Loops:

Not learned Learned **Proficient**

One-Dimensional Arrays:

Not learned Learned Proficient

Two-Dimensional Arrays:

Not learned Learned Proficient

Character Arrays:

Not learned Learned Proficient

Strings:

Not learned **Learned** Proficient

Custom Functions:

Not learned **Learned** Proficient

Struct:

Not learned Learned Proficient

Files and Basic I/O:

Not learned Learned Proficient

STL Templates:

Not learned Learned Proficient

Algorithms and Data Structures (Entry Level)

Enumeration:

Not learned **Learned** Proficient

Simulation:

Not learned Learned Proficient

Bubble Sort:

Not learned Learned Proficient

Quick Sort:

Not learned Learned Proficient

Queues:

Not learned Learned Proficient

Algorithms and Data Structures (Advanced Level)

Shortest Path	Not learned	Learned	Proficient
Single-Source Second Shortest Path	Not learned	Learned	Proficient
Minimum Spanning Tree (MST)	Not learned	Learned	Proficient
Second Minimum Spanning Tree	Not learned	Learned	Proficient
Topological Sorting	Not learned	Learned	Proficient
Eulerian Path and Eulerian Circuit	Not learned	Learned	Proficient
Bipartite Graph Determination	Not learned	Learned	Proficient
Strongly Connected Components (SCC)	Not learned	Learned	Proficient
Articulation Points & Bridges	Not learned	Learned	Proficient
Union-Find (Disjoint Set Union)	Not learned	Learned	Proficient
Fenwick Tree (Binary Indexed Tree)	Not learned	Learned	Proficient
Segment Tree	Not learned	Learned	Proficient
Splay Tree	Not learned	Learned	Proficient
Treap	Not learned	Learned	Proficient
Center, Diameter and Centroid of a Tree	Not learned	Learned	Proficient
DFS Order and Euler Tour of a Tree	Not learned	Learned	Proficient
Tree Difference, Subtree Sum & Doubling	Not learned	Learned	Proficient
Lowest Common Ancestor (LCA)	Not learned	Learned	Proficient
Discretisation	Not learned	Learned	Proficient
Divide and Conquer	Not learned	Learned	Proficient
Radix Sort	Not learned	Learned	Proficient
Tree DP	Not learned	Learned	Proficient
State Compression DP	Not learned	Learned	Proficient
Common DP Optimisations	Not learned	Learned	Proficient

Euler's Theorem & Euler's Totient Function

Bézout's Identity

Modular Multiplicative Inverse

Extended Euclidean Algorithm

Chinese Remainder Theorem

Permutations & Combinations on Multisets

Derangements & Circular Permutations

Inclusion-Exclusion Principle

Catalan Numbers

Vectors and Matrices

Gaussian Elimination

String Matching (KMP)

Not learned Learned Proficient

Not learned Learned Proficient

Not learned Learned Proficient

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Not learned Learned Proficient

Stacks:

Not learned Learned Proficient

Linked Lists:

Not learned Learned Proficient

Trees and Binary Trees:

Not learned Learned Proficient

Binary Search:

Not learned Learned Proficient

Binary Lifting

Not learned Learned Proficient

High-Precision Arithmetic

Not learned Learned Proficient

Recursion

Not learned Learned Proficient

Greedy Algorithms

Not learned Learned Proficient

Depth-First Search (DFS)

Not learned Learned Proficient

Breadth-First Search (BFS)

Not learned Learned Proficient

Basics of Dynamic Programming (DP)

Not learned Learned Proficient

1D Dynamic Programming

Not learned Learned Proficient

Knapsack Problems

Not learned Learned Proficient

Interval DP

Not learned Learned Proficient

Euclidean Algorithm

Not learned Learned Proficient

Prime Sieves: Sieve of Eratosthenes

& Linear Sieve

Not learned Learned Proficient

Sets

Not learned Learned Proficient

Addition Principle (Rule of Sum) &

Multiplication Principle (Rule of Multiplication)

Not learned Learned Proficient

Permutations and Combinations

Not learned Learned Proficient

STL

Not learned Learned Proficient

Prefix Sum

Not learned Learned Proficient

Difference Array

Not learned Learned Proficient